Skills Summary

Graphing Proportional Relationships

Use this reference while completing your worksheet or when studying.

The test for proportional. The graph is a *straight line* and it passes through the *origin* $(0, 0)$. Both halves are needed: a straight line that misses the origin is linear but not proportional.

The constant of proportionality (unit rate). $k = \dfrac{y}{x}$ at any point of the line except the origin. Equivalently k is the rise over the run.

The equation. $y = kx$ — no added constant, because the line starts at $(0, 0)$. Reading a point off the graph gives k; the equation then answers questions the drawn axis is too short for.

1. From a table to a graph through the origin

Plot each row as a point (x, y) and rule one straight line through them. If the relationship is proportional the line hits $(0, 0)$, so $(0, 0)$ is always a legal extra point to plot. The unit rate is then the y-value at $x = 1$.

Example: A table gives 3 hours of work paying 12 dollars. Plot it with the origin and find the unit rate.
Step 1. Pay with no work done is 0 dollars, so $(0, 0)$ belongs on the graph; with $(3, 12)$ that is enough to rule the line and read its steepness.
Step 2. $k = \dfrac{12 - 0}{3 - 0} = \dfrac{12}{3}$. The line through the origin rises **4** dollars for every hour.

Try it: A table gives 4 kilograms of apples costing 10 dollars. What is the cost of one kilogram, in dollars?

check: **2.5**

2. Reading the unit rate off a drawn graph

Find a point where the line crosses a corner of the grid, then divide: the y-axis quantity goes on top. Dividing the wrong way round gives the reciprocal — a small number where a large one belongs.

Example: A line through the origin passes through $(6, 9)$. Find its unit rate.
Step 1. The line already passes through the origin, so one clean lattice point is all the information a unit rate needs.
Step 2. $k = \dfrac{9}{6}$, which is the rise 9 over the run 6. The unit rate is **1.5** per unit of x.

Try it: A different line through the origin passes through $(8, 28)$. What is its unit rate?

check: **3.5**

3. Using the equation to predict

Once k is known, $y = kx$ answers both directions: substitute an x to predict a y, or set y to a target value and solve for x. This is how you answer a question whose numbers run past the end of the drawn axis.

Example: A graph through the origin has unit rate $k = 4$. Predict y when $x = 7$, a value past the right-hand end of the drawn axis.
Step 1. The graph stops before $x = 7$, so read k off the graph and let the equation $y = kx$ carry the prediction.
Step 2. $y = 4 \times 7$. The predicted output is **28**.

Try it: A line through the origin has unit rate 6. Solve $6x = 45$ to find the x that gives an output of 45.

check: **7.5**

(!) Watch out: a straight line is not enough. Check the origin before calling a relationship proportional, and check that every $y \div x$ in a table gives the same number. Dividing x by y is the other frequent slip: the unit rate answers "how much y per one x", so y is on top.